



This thesis establishes a computational framework for the design of membrane-targeted therapeutic peptides. First, metadynamics maps peptide-membrane free-energy landscapes, identifying the transverse hydrophobic dipole moment as the key descriptor governing spontaneous insertion into pathological envelopes. Next, a resource-efficient quantum algorithm predicts interface-driven peptide folding without escalating hardware demands. Finally, these insights inform a Z-score normalized Hamiltonian for the de novo design of environment-conditioned  $\alpha$ -helical sequences. Solved via hybrid quantum algorithms, this pipeline effectively bridges biophysics and the discovery of selective peptidomimetics.